FuMOGAE: A Fuzzy Multi-Omics Graph AutoEncoder for Breast Cancer Subtype Classification

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1 Supplementary Methods

1.1 Dataset Descriptions

A brief description of each data modality is given below.

1. **Clinical (CLN) :** Clinical data comprise clinical and pathological attributes such as demographic information, cancer history, tumour stage, receptor status, and treatment details. Numerical variables like age are normalized using min–max scaling, while categorical variables are encoded as integers to ensure uniform representation across samples.
2. **Copy Number Aberrations (CNA) :** CNA profile describe large scale genomic gain and loses derived from GISTIC 2-thresholded data. Each gene is assigned a discrete value indicating its copy number state:-2 for homozygous deletion,-1 for single-copy loss,0 for normal diploid state,+1 for low-level amplification,+2 for high-level amplification. These values reflect the extent of genomic instability present in each sample.
3. **DNA Methylation (DNA) :** Methylation measurement were obtained from the iilumina human methylation 27 and human methylation 450 arrays. Each feature represents a β-values that quantifies methylation intensity at individual CpG sites. These values capture epigenetic modifications that influence gene regulation and contribute to tumour development and progression.
4. **microRNA (miRNA) Expression :** The miRNA dataset records expression level of mature microRNAs,which serve as key post transcriptional regulator of gene expression. Data from iiluminaGA and iiluminaHiSeq platform were harmonized to ensure comparability, providing insight into regulatory mechanism affecting mRNA stability and translation.
5. **Gene Expression (GSE) :** Gene expression profiles represent transcriptomic activity measured through microarray or RNA-seq platforms. Expression values were normalized and filtered to retain high-variance genes, ensuring that informative features reflecting transcriptional differences across subtypes are preserved.

6. **Mutation :** Mutation data encode somatic alterations [2] in cancer-related genes. Each feature represents the presence or absence of a mutation, providing insights into genomic instability and oncogenic pathways underlying breast cancer subtypes.
7. **Co-expression (COEXP) :** Co-expression features were generated using the Weighted Gene Co-expression Network Analysis (WGCNA) tool [1], which identifies gene modules based on pairwise expression correlations across patients. Each module is summarized by its eigengene, representing the principal component of expression within the module and capturing coordinated gene regulation patterns.

1.2 Computational Environment

All experiments were performed on a workstation with **NVIDIA RTX A100** (48 GB VRAM), **Intel Xeon Gold 6248R** CPU (3.0 GHz), and **256 GB RAM**. Software stack: **Python 3.10**, **PyTorch 2.0**, **PyTorch Geometric 2.3**, **scikit-learn 1.3.1**, **XGBoost 1.7**. End-to-end training time per outer fold: ∼8–10 minutes (single-modality) and ∼30–35 minutes (multimodal fusion), excluding figure generation.

1.3 Computational Complexity

The computational complexity of the proposed **FuMOGAE** framework can be analyzed across four stages: graph construction, graph autoencoder (GAE) training, classifier training, and Choquet–Sugeno fuzzy fusion.

Graph Construction. For each modality, a patient–patient Pearson correlation matrix is computed, requiring $O(N^2 d_m)$ operations for N patients and d_m features. This is followed by k-nearest neighbor selection and thresholding with an additional $O(N^2)$ cost; the final sparse graph stores approximately $E_m \approx N k_m$ edges. Thus, graph construction is dominated by the $O(N^2 d_m)$ term, especially for high-dimensional omics data.

GAE Training. The encoder uses two graph convolution layers and a linear decoder, giving a per-epoch complexity of

$$O\big(E_m(d_m + h) + N(d_m h + h^2)\big)\,,$$

where h is the latent dimension. Over T epochs, the cost becomes

$$O\big(T\,[E_m(d_m + h) + N(d_m h + h^2)]\big)\,,$$

repeated independently for each modality.

Classifier Training. With embeddings obtained, nested cross-validation trains classifiers and estimates modality weights. For an FFNN with input dimension d and hidden size H, complexity is $O(T_{\text{ff}}\, N(dH+H^2))$ per run, where T_{ff} is epochs. Random Forest scales as $O(T_{\text{trees}}\, Nd \log N)$, XGBoost as $O(T_{\text{boost}}\, Nd \log N)$, and RBF-SVM can reach $O(N^3)$ in the worst case. An $F_{\text{outer}} \times F_{\text{inner}}$ nested CV scales these costs multiplicatively.

For classification, five base classifiers were employed: Feed-Forward Neural Network (FFNN), Multi-Layer Perceptron (MLP), XGBoost, Random Forest (RF), and Support Vector Machine (SVM).

– **FFNN:** Three hidden layers (256, 128, 64) with ReLU activations, dropout (p = 0.4), and AdamW[3] optimizer (learning rate = 0.001, 200 epochs). Softmax was used in the output layer for class probabilities.

– **MLP:** Trained with Adam[3] optimizer, adaptive learning-rate scheduling, and 200 iterations.

– **XGBoost:** 600 trees, max_depth = 6, learning_rate = 0.03, subsample = 0.9, colsample_bytree = 0.8, objective = `multi:softprob`.

– **Random Forest:** 400 trees, balanced subsample; baseline RF with `n_estimators = 100` (fixed via grid search).

– **SVM:** RBF kernel with C = 2.0, γ = scale, class-weight balancing.

Choquet–Sugeno Fuzzy Fusion. Fusion aggregates modality-specific probabilities per sample and class. Sorting and aggregation yield $O(N\, C\, m_c \log m_c)$ for N samples, C classes, and m_c modalities. This is negligible compared to graph construction and classifier training.

Overall. The computational burden is dominated by $O(N^2 d_m)$ for graph construction and $O(T[E_m(d_m + h) + N(d_m h + h^2)])$ for GAE training, followed by classifier training under nested CV, while the fuzzy fusion overhead remains minimal.

2 Supplementary Tables

Tables S1–S4 summarize the detailed quantitative results for TCGA, METABRIC, and Combined datasets. They report Macro F1-score, Weighted F1-score, and Accuracy for single modalities (CLI, CNA, COE, EXP, DNA, MIR, MUT) and for various classifiers (FFNN, MLP, RF, SVM, XGBoost).

Representative results: - On TCGA data, ChoqSugFuzGAE achieved a Weighted F1-score of 0.8077 (CLI), outperforming MOGONET (0.7036) and MLP (0.7112). - On METABRIC, the macro F1 improved to 0.8165 (CLI), showing superior consistency across datasets. - Integration results (Table 4) show the highest macro F1 of 1.00 under RF and XGBoost classifiers. - Gene expression–only comparison (Table S2) demonstrates consistent superiority of ChoqSugFuzGAE over DeepCC, GCNC-COE, and GCNC-PPI across all datasets.

Table 1: **Single model results on TCGA data** – Weighted F1-score and Accuracy for each model with a single datatype. [Datatypes: CLI (clinical), CNA (copy number aberration), COE (coexpression), EXP (gene expression), DNA (DNA sequence features), MIR (microRNA expression), MUT (mutation). MLP stands for Multi-layer Perceptron; ChoqSugFuzGAE and ChoqSugFuzGAE- are our proposed models.]

Weighted F1-scores							
Method	**CLI**	**CNA**	**EXP**	**DNA**	**MIR**	**MUT**	**COE**
FuMOGAE	0.8077 ± 0.2216	0.3905 ± 0.0586	0.5915 ± 0.0418	0.5423 ± 0.1405	0.7551 ± 0.0357	0.3375 ± 0.0301	0.4087 ± 0.0423
FuMOGAE-	0.6542 ± 0.1022	0.4414 ± 0.0512	0.5820 ± 0.0497	0.6138 ± 0.0323	0.6812 ± 0.0567	0.2922 ± 0.0349	0.3829 ± 0.0417
MOGONET	0.7036 ± 0.0579	0.4414 ± 0.0442	0.5885 ± 0.0435	0.6229 ± 0.0366	0.7134 ± 0.0333	0.2596 ± 0.0383	0.3604 ± 0.0364
MLP	0.7112 ± 0.0485	0.3850 ± 0.0581	0.5484 ± 0.0380	0.6288 ± 0.0249	0.6850 ± 0.0411	0.3065 ± 0.0289	0.3977 ± 0.0416
Accuracies							
FuMOGAE	0.8350 ± 0.1857	0.4038 ± 0.0551	0.5954 ± 0.0410	0.5651 ± 0.1236	0.7550 ± 0.0358	0.3591 ± 0.0406	0.4540 ± 0.0505
FuMOGAE-	0.6542 ± 0.1022	0.4501 ± 0.0563	0.5922 ± 0.0435	0.6018 ± 0.0303	0.6824 ± 0.0553	0.2920 ± 0.0306	0.4046 ± 0.0437
MOGONET	0.7159 ± 0.0515	0.4916 ± 0.0426	0.6106 ± 0.0397	0.6353 ± 0.0394	0.7191 ± 0.0344	0.3966 ± 0.0145	0.4548 ± 0.0358
MLP	0.7119 ± 0.0476	0.3909 ± 0.0632	0.5523 ± 0.0392	0.6297 ± 0.0255	0.6871 ± 0.0398	0.3041 ± 0.0327	0.3967 ± 0.0435

Table 2: **Single model results on METABRIC data** – Macro F1-score, Weighted F1-score, and Accuracy for each model with a single datatype. [CLI: clinical, CNA: copy number aberration, COE: coexpression, EXP: gene expression, MUT: mutation. MLP: Multi-layer Perceptron.]

Macro F1-scores

Method	**CLI**	**CNA**	**COE**	**EXP**	**MUT**
FuMOGAE-	$\mathbf{0.6269 \pm 0.1074}$	0.3887 ± 0.0387	0.3385 ± 0.0434	0.5457 ± 0.0326	0.2472 ± 0.0231
FuMOGAE	$\mathbf{0.8165 \pm 0.2075}$	0.3625 ± 0.0672	0.3545 ± 0.0506	0.5445 ± 0.0553	0.2972 ± 0.0523
MOGONET	0.6620 ± 0.0493	0.3637 ± 0.0560	0.3021 ± 0.0631	0.5521 ± 0.0449	0.1484 ± 0.0371
MLP	$\mathbf{0.6768 \pm 0.0595}$	0.3476 ± 0.0476	0.3436 ± 0.0410	0.5246 ± 0.0496	0.2484 ± 0.0367

Weighted F1-scores

Method	**CLI**	**CNA**	**COE**	**EXP**	**MUT**
FuMOGAE-	$\mathbf{0.6569 \pm 0.0978}$	0.4363 ± 0.0534	0.3872 ± 0.0438	0.5733 ± 0.0350	0.3018 ± 0.0277
FuMOGAE	$\mathbf{0.8359 \pm 0.1952}$	0.3862 ± 0.0830	0.4062 ± 0.0479	0.5782 ± 0.0541	0.3621 ± 0.0457
MOGONET	0.6899 ± 0.0489	0.4501 ± 0.0446	0.3717 ± 0.0452	0.5892 ± 0.0341	0.2580 ± 0.0373
MLP	$\mathbf{0.7112 \pm 0.0485}$	0.3850 ± 0.0581	0.3977 ± 0.0416	0.5484 ± 0.0380	0.3065 ± 0.0289

Accuracies

Method	**CLI**	**CNA**	**COE**	**EXP**	**MUT**
FuMOGAE-	$\mathbf{0.6832 \pm 0.0647}$	0.4445 ± 0.0643	0.4238 ± 0.0364	0.5778 ± 0.0306	0.3017 ± 0.0210
FuMOGAE	$\mathbf{0.8542 \pm 0.1703}$	0.4085 ± 0.0625	0.4493 ± 0.0375	0.5818 ± 0.0513	0.3759 ± 0.0433
MOGONET	0.6984 ± 0.0455	0.4996 ± 0.0377	0.4637 ± 0.0266	0.6129 ± 0.0290	0.3926 ± 0.0163
MLP	$\mathbf{0.7119 \pm 0.0476}$	0.3909 ± 0.0632	0.3967 ± 0.0435	0.5523 ± 0.0392	0.3041 ± 0.0327

Table 3: **Single model results on the COMBINED data (TCGA+METABRIC)** – Macro F1-score, Weighted F1-score, and Accuracy for each model with a single datatype. [CLI: clinical, EXP: gene expression, MUT: mutation. MLP: Multi-layer Perceptron.]

Macro F1-scores

Method	**CLI**	**EXP**	**MUT**
FuMOGAE-	$\mathbf{0.6085 \pm 0.0789}$	0.5379 ± 0.0431	0.2416 ± 0.0641
FuMOGAE	$\mathbf{0.8439 \pm 0.2237}$	0.5635 ± 0.0424	0.2827 ± 0.0510
MOGONET	0.6620 ± 0.0493	0.5418 ± 0.0535	0.1501 ± 0.0331
MLP	$\mathbf{0.6637 \pm 0.0605}$	0.5044 ± 0.0454	0.2438 ± 0.0427

Weighted F1-scores

Method	**CLI**	**EXP**	**MUT**
FuMOGAE-	$\mathbf{0.6447 \pm 0.0720}$	0.5579 ± 0.0406	0.2974 ± 0.0617
FuMOGAE	$\mathbf{0.8539 \pm 0.2180}$	0.5888 ± 0.0460	0.3531 ± 0.0439
MOGONET	0.6899 ± 0.0489	0.5815 ± 0.0343	0.2553 ± 0.0385
MLP	$\mathbf{0.6980 \pm 0.0575}$	0.5262 ± 0.0376	0.3075 ± 0.0497

Accuracies

Method	**CLI**	**EXP**	**MUT**
FuMOGAE-	$\mathbf{0.6713 \pm 0.0497}$	0.5571 ± 0.0444	0.2970 ± 0.0633
FuMOGAE	$\mathbf{0.8784 \pm 0.1784}$	0.5866 ± 0.0427	0.3815 ± 0.0389
MOGONET	0.6984 ± 0.0455	0.6026 ± 0.0295	0.3894 ± 0.0350
MLP	$\mathbf{0.6991 \pm 0.0568}$	0.5300 ± 0.0351	0.3073 ± 0.0557

Table 4: **FuMOGAE performance with different integration methods.** Minimum, median, and maximum macro F1 across TCGA, METABRIC, and Combined datasets. [FuMOGAE-: without raw feature integration]

Setup	**TCGA**	**METABRIC**	**COMBINED**
FuMOGAE (FuMOGAE FFNN)	0.67, 0.88, **0.96**	0.93, **0.98**, **1.00**	0.52, **0.94**, **1.00**
FuMOGAE (FuMOGAE MLP)	0.80, **0.84**, **0.91**	0.81, 0.88, **0.93**	0.85, **0.90**, **0.97**
FuMOGAE- (FuMOGAE-FFNN)	0.87, **0.93**, **0.95**	**0.93**, **0.98**, **1.00**	0.64, **0.97**, **1.00**
FuMOGAE- (FuMOGAE-MLP)	0.80, **0.88**, 0.91	0.83, **0.90**, **0.94**	**0.93**, **0.97**, **1.00**
FuMOGAE (RF)	**0.85**, **0.90**, **0.97**	**0.94**, **0.97**, **1.00**	**0.98**, **1.00**, **1.00**
FuMOGAE- (RF)	**0.90**, **0.94**, **0.99**	**0.95**, **0.99**, **1.00**	**0.98**, **1.00**, **1.00**
FuMOGAE (SVM)	**0.82**, **0.88**, 0.93	**0.86**, **0.96**, **0.98**	**0.92**, **0.97**, **1.00**
FuMOGAE- (SVM)	**0.87**, **0.92**, **0.98**	**0.92**, **0.96**, **1.00**	**0.93**, **0.97**, **1.00**
FuMOGAE (XGBoost)	**0.91**, **0.94**, **0.99**	**0.91**, **0.93**, **0.97**	**0.98**, **1.00**, **1.00**
FuMOGAE- (XGBoost)	**0.80**, **0.89**, 0.92	**0.94**, **0.96**, **0.98**	**0.98**, **1.00**, **1.00**

Table 5: Parameter Settings: Hyperparameter configurations used in FuMOGAE experiments (default values unless otherwise noted).

Component	Parameter	Value
5*GAE Encoder (per modality)	Layers	GCN(256) $\rightarrow$ GCN(128)
	Activation	ReLU
	Dropout	0.40
	Learning rate	0.010
	Epochs	10
4*Graph Construction	Similarity metric	Pearson correlation
	k-NN (CLN / others)	10 / 5
	Thresholds τ	CLN: 0.30; CNA: 0.60; GSE: 0.30; DNA: 0.30; miRNA: 0.35; MUT: 0.25; COEXP: 0.40
5*Classifier (FFNN)	Hidden sizes	[256, 128, 64]
	Dropout	0.40
	Optimizer	AdamW
	Learning rate	0.001
	Epochs / Batch size	200 / 128
4*Other Classifiers	RF	400 trees, class_weight=balanced_subsample
	XGBoost	600 trees, depth=6, η=0.03, subsample=0.9, colsample=
	SVM (RBF)	C=2.0, γ=scale, class_weight=balanced
	MLP (sklearn)	max_iter=200, adaptive LR
2*Training Protocol	CV	10-fold outer; 5-fold inner for weights w_m
	Imbalance handling	SMOTE (train-fold only)
2*Fuzzy Fusion	Reliability α	{0.3, 0.5, 0.7, 0.9}
	Aggregation	α Choquet + $(1-\alpha)$ Sugeno

Table 6: Evaluation Metrics: Metric definitions used for evaluation. TP, FP, TN, FN are class-conditioned counts; macro/weighted averages follow scikit-learn conventions.

Metric	Definition
Accuracy	$\dfrac{TP + TN}{TP + TN + FP + FN}$
Precision (per class c)	$\dfrac{TP_c}{TP_c + FP_c}$
Recall (per class c)	$\dfrac{TP_c}{TP_c + FN_c}$
F1-score (per class c)	$\dfrac{2\,\mathrm{Prec}_c\,\mathrm{Rec}_c}{\mathrm{Prec}_c + \mathrm{Rec}_c}$
Macro-F1	Unweighted mean of per-class F1-scores
Weighted-F1	Support-weighted mean of per-class F1-scores
AUC (per class c)	Area under ROC of c vs. rest (one-vs-rest); macro-averaged across classes

3 Supplementary Algorithm: FuMOGAE

3.1 Algorithmic Framework

Algorithm 1 Two-Stage Hybrid Feature Selection using Boruta + mRMR

Require: Feature matrix $X \in R^{n \times d}$, labels y
Ensure: Final selected feature subset $\mathcal{F}_{\text{mRMR}}$
Initialize: $\mathcal{F}_B \leftarrow \emptyset$, $\mathcal{F}_{\text{mRMR}} \leftarrow \emptyset$

Stage 1 – Boruta:
for each feature f_i in X **do**
    Generate shadow features $\tilde{f}_i$ by permutation
    Compute Random Forest importance $I(f_i)$
    if $I(f_i) > \max(I_{\tilde{X}})$ **then**
        Add f_i to $\mathcal{F}_B$ ▷ Confirmed relevant
    end if
end for

Stage 2 – mRMR:
Initialize selected set $S \leftarrow \emptyset$
while $|S| < k$ **do**
    for each $f_i \in \mathcal{F}_B \setminus S$ **do**
        Compute:

$$\text{mRMR}(f_i) = I(f_i; y) - \frac{1}{|S|} \sum_{f_j \in S} I(f_i; f_j)$$

    end for
    Select $f^* = \arg\max_{f_i} \text{mRMR}(f_i)$
    Add f^* to S
end while
$\mathcal{F}_{\text{mRMR}} \leftarrow S$
return $\mathcal{F}_{\text{mRMR}}$

Algorithm 2 Choquet–Sugeno Fuzzy Fusion with GAE-based Embeddings for Breast Cancer Subtype Classification

Require: Preprocessed modality CSVs: Clinical, CNA, GSE, DNA, miRNA, Mutation, COEXP
Ensure: Subtype-wise and overall performance metrics, confusion matrices, fusion results

Step 1: Data Loading
Read preprocessed CSVs for all modalities.
Extract features (exclude PatientID and label).
Extract labels (last column of Clinical).

Step 2: Feature Selection Outputs
Save modality-specific features as `selected_*.npy/.csv`.
Save labels as `labels.npy`.

Step 3: Graph Construction
for each modality $M \in \{\text{Clinical}, \text{CNA}, \text{GSE}, \text{DNA}, \text{miRNA}, \text{Mutation}, \text{COEXP}\}$ **do**
    Compute Pearson correlation matrix S across patients.
    Select top-k neighbors for each node using $|S|$.
    Apply modality-specific threshold τ_M to prune weak edges.
    Build undirected weighted adjacency matrix A.
    Form PyG graph $\mathcal{G}_M = (X, A)$ with features X.
end for

Step 4: Graph Autoencoder Training
for each modality graph $\mathcal{G}_M$ **do**
    Encode features with 2-layer GCN encoder $\phi(X, A) \to Z$.
    Decode with linear layer $\psi(Z) \to \hat{X}$.
    Train by minimizing MSE: $\mathcal{L} = \|X - \hat{X}\|^2$.
    Save latent embeddings Z_M.
end for

Step 5: Embedding Integration
Horizontally concatenate $[Z_M, X_M]$ for each modality to form `combined_M`.
Save both latent (Z_M) and combined arrays.

Step 6: Model Training & Evaluation
for mode $\in \{\text{EMB}, \text{RAWEMB}\}$ **do**
    for each modality combination C **do**
        for classifier $\in \{\text{FFNN}, \text{SVM}, \text{RF}, \text{MLP}, \text{XGB}\}$ **do**
            Perform 10-fold Stratified CV:
            for outer fold $f = 1 \to 10$ **do**
                Split train/test sets.
                for each modality $m \in C$ **do**
                    Run inner 5-fold CV to compute reliability weight w_m.
                    Train classifier with StandardScaler + SMOTE.
                    Obtain predicted probabilities p_m on outer test set.
                end for
                Fuse probabilities via Choquet & Sugeno integrals:

$$p^{\text{fused}}(c) = \alpha\, C(\{p_m(c)\}, \{w_m\}) + (1 - \alpha)\, S(\{p_m(c)\}, \{w_m\})$$

                Predict label $\hat{y} = \arg\max_c\, p^{\text{fused}}(c)$.
                Compute fold metrics (Accuracy, Macro-F1, Weighted-F1).
            end for
            Aggregate results across folds (mean ± SD).
            Save subtype-wise metrics, confusion matrix, and plots.
        end for
    end for
end for

Output:
Per-fold CSVs, subtype metrics tables, confusion matrix heatmaps, and global summaries of Accuracy, Macro-F1, Weighted-F1.

Algorithm 3 Pearson-kNN Patient–Patient Graph Construction (Per Modality)

Require: Row-aligned matrices (rows = patients): $X_{\text{CLN}}, X_{\text{CNA}}, X_{\text{GSE}}, X_{\text{DNA}}, X_{\text{miRNA}}, X_{\text{MUT}}, X_{\text{COEXP}}$
Require: Modality set $\mathcal{M} = \{\text{CLN}, \text{CNA}, \text{GSE}, \text{DNA}, \text{miRNA}, \text{MUT}, \text{COEXP}\}$
Require: **kNN values**: k_{CLN}=10, k_{CNA}=5, k_{GSE}=5, k_{DNA}=5, k_{miRNA}=5, k_{MUT}=5, k_{COEXP}=5
Require: **Thresholds** τ: τ_{CLN}=0.30, τ_{CNA}=0.60, τ_{GSE}=0.30, τ_{DNA}=0.30, τ_{miRNA}=0.35, τ_{MUT}=0.25, τ_{COEXP}=0.40
Ensure: PyG graphs with signed edge weights: $\mathcal{G}_{\text{CLN}}, \mathcal{G}_{\text{CNA}}, \mathcal{G}_{\text{GSE}}, \mathcal{G}_{\text{DNA}}, \mathcal{G}_{\text{miRNA}}, \mathcal{G}_{\text{MUT}}, \mathcal{G}_{\text{COEXP}}$

function BUILDGRAPH(X, k, τ)
    $N \leftarrow$ #patients (rows of X)
    $k \leftarrow \min(\max(1, k), \max(1, N-1))$
    $S \leftarrow$ corrcoef(X); $\forall i : S_{ii} \leftarrow 0$; $S \leftarrow$ nan_to_num(S); $S \leftarrow$ clip($S, -1, 1$)
    $R \leftarrow |S|$ ▷ select neighbors by absolute correlation; keep signed weights from S
    Initialize triplets $(\mathbf{r}, \mathbf{c}, \mathbf{v}) \leftarrow \emptyset$
    for $i \leftarrow 1$ **to** N **do**
        $\mathcal{N}_i \leftarrow$ indices of top-k entries of $R_{i:}$ (descending)
        for $j \in \mathcal{N}_i$ **do**
            if $|S_{ij}| \geq \tau$ **then**
                append (i, j, S_{ij}) to $(\mathbf{r}, \mathbf{c}, \mathbf{v})$
            end if
        end for
    end for
    $A \leftarrow$ csr_matrix($(\mathbf{v}), (\mathbf{r}, \mathbf{c}), (N, N)$)
    $A \leftarrow A$.maximum($A^\top$) ▷ elementwise max; preserves the larger *signed* entry
    (`edge_index`, `edge_weight`) $\leftarrow$ from_scipy_sparse_matrix(A)
    return Data(x:=torch.from_numpy(X), `edge_index`, `edge_weight`)
end function
for each $M \in \mathcal{M}$ **do**
    $(X_M, k_M, \tau_M) \leftarrow$
     if M=CLN: $(X_{\text{CLN}}, 10, 0.30)$; **elif** M=CNA: $(X_{\text{CNA}}, 5, 0.60)$; **elif** M=GSE: $(X_{\text{GSE}}, 5, 0.30)$;
     elif M=DNA: $(X_{\text{DNA}}, 5, 0.30)$; **elif** M=miRNA: $(X_{\text{miRNA}}, 5, 0.35)$;
     elif M=MUT: $(X_{\text{MUT}}, 5, 0.25)$; **elif** M=COEXP: $(X_{\text{COEXP}}, 5, 0.40)$
    $\mathcal{G}_M \leftarrow$ BUILDGRAPH(X_M, k_M, τ_M)
end for

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